HW5

Question B)

But now we have $\max\{S_1, S_2\}$

$= \max\{S_1 - S_2, 0\} + S_2$

S_2: value at maturity $S_2(T)$, but now has value $S_2(0)$

$dS_1 = \mu_1 S_1 dt + \sigma S_1 dW_1$

$dS_2 = \mu_2 S_2 dt + \sigma S_2 dW_2$

$f(S_1, S_2, t)$ Itô's lemma

$df = f_t dt + f_1 dS_1 + f_2 dS_2 + \frac{1}{2} f_{11} dS_1^2 + f_{12} dS_1 dS_2 + \frac{1}{2} f_{22} dS_2^2$

$dt\,dt = 0,\ dt\,dW_i = 0,\ dW_i\,dW_i = dt,\ dW_1\,dW_2 = 0$

⇓ substituting $(dS_i)^2$ terms ...

$df = f_t dt + f_1 dS_1 + f_2 dS_2 + \frac{1}{2}\left(\sigma^2 S_2^2 + \sigma^2 S_1^2\right) dt$

$\Pi = f(S_1, S_2, t) - \Delta_1 S_1 - \Delta_2 S_2$

$d\Pi = f_t dt + f_1 dS_1 + f_2 dS_2 + \frac{1}{2}\left(\sigma^2 S_2^2 f_{22}'' + \sigma^2 S_1^2 f_{11}''\right) dt - \Delta_1 dS_1 - \Delta_2 dS_2$

wrt dS_1, dS_2 should vanish

$\Rightarrow \Delta_1 = f_1', \quad \Delta_2 = f_2'$

$\Pi = f(S_1, S_2, t) - f_1' dS_1 - f_2' dS_2$

$d\Pi = \left(f_t' + \frac{1}{2}\sigma^2 S_2^2 + \frac{1}{2}\sigma^2 S_1^2\right) dt$

$= r\Pi\, dt$

$f_t' + \frac{1}{2}\sigma^2\left(S_1^2 f_{11}'' + S_2^2 f_{22}''\right) = r\left(f - f_1' S_1 - f_2' S_2\right)$

↑

PDE with these boundary & terminal conditions

$f(S_1, S_2, T) = \max\{S_1, S_2\}$

$f(0, S_2, t) = S_2(t)$

$f(S_1, 0, t) = S_1(t)$

$f(S_1, S_2, 0) = S_1(0) = 100$

Question a)

Black Scholes for $\max\{S(T) - K, 0\}$ has

$S(0) N(d_1) - K N(d_2) e^{-rT}$

where $d_1, d_2 = \dfrac{\log\left(\frac{S}{K}\right) \pm T\left(r + \frac{\sigma^2}{2}\right)}{\sigma\sqrt{T}}$

So applying that to $\max\{S_1 - S_2, 0\} + S_2$ would be using a modified Black Scholes w/ $K = S_2$ and no discounting for $S_2(0)$

$S_1(0) N(d_1) - S_2(0) N(d_2) + S_2(0)$

$d_1, d_2 = \dfrac{\log\left(\frac{S_1(0)}{S_2(0)}\right) \pm T\left(r + \frac{1}{2}\hat{\sigma}^2\right)}{\hat{\sigma}\sqrt{T}} = \dfrac{\log\left(\frac{S_1(0)}{S_2(0)}\right) \pm T\left(r + \sigma^2\right)}{\sigma\sqrt{2T}}$

But $\hat{\sigma}$ here is different from the standard BSM because the ratio $\frac{S_1(0)}{S_2(0)}$ will depend on both stocks

↳ variances add up $\hat{\sigma}^2 = 2\sigma^2$ both stocks same σ

$\hat{\sigma} = \sqrt{2}\,\sigma$

Question c)

analytic sol

$$\max\{S_1, S_2\} = \max\{S_1 - S_2, 0\} + S_2$$

↙ use BSM for S_1 w/ $K = S_2$

$$S_2 + S_1 * N(d_1) - S_2 * N(d_2)$$

$$d_1, d_2 = \frac{\log\left(\frac{S_1}{S_2}\right) + \frac{1}{2}\sigma^2 \cdot 2T}{\sqrt{2}\sigma\sqrt{T}}$$

* But volatility is $2\sigma^2$ instead
  $\Rightarrow \sigma \leftarrow \sqrt{2}\sigma$

* No discounting needed for S_2 (instead of K)

Finite Diff Method

$$x = \log\left(\frac{S_1}{S_0}\right) = \log\left(\frac{S_2}{S_0}\right)$$

$$dV/dx = \frac{1}{2}[V[j+1] - V[j-1]]$$

$$d^2V/dx^2 = V[j-1] - 2V[j] + V[j+1]$$

using central method

Now the PDE

$$V_t + \frac{1}{2}\sigma^2 S_1^2 V_{11}'' + r S_1 V_1' + \frac{1}{2}\sigma^2 S_2^2 V_{22}'' + r S_2 V_2' - rV = 0$$

look at derivatives wrt S_1 (or S_2)

$$\mathcal{L} = \frac{1}{2}\sigma^2 S_1 \frac{\partial^2 V}{\partial S_1^2} + r S_1 \frac{\partial V}{\partial S_1}$$

Substitute $x = \log\left(\frac{S_1}{S_0}\right)$ $dx = \frac{1}{S_1} dS_1$

$$\mathcal{L} = \frac{1}{2}\sigma^2\left(\frac{\partial^2 V}{\partial x^2} - \frac{\partial V}{\partial x}\right) + r\frac{\partial V}{\partial x}$$

For code get_d1 $\cong \frac{\partial V}{\partial x}$ using central

get_d2 $\cong \frac{\partial^2 V}{\partial x^2}$ using central

get_L $\cong \mathcal{L}$ using ↗